



STUDENT PERFORMANCE G.C.E A\L EXAM 2020 SRI LANKA

GROUP 4

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INTRODUCTION

The GCE Advanced Level (A/L) 2020 Sri Lanka Student Performance dataset contains anonymized academic records of students who sat for the national A/L examination in 2020. It includes subject-wise marks, overall Z-scores, streams of study and student information. This dataset is useful for analyzing student performance patterns and identifying factors that influence academic achievement in Sri Lanka.

VARIABLES OF THE DATASET

- Index - unique identifier
- Stream - Subjects stream students faced in the exam
- z score - z scores of the students
- sub1 - name of the 1st subject
- sub2 - name of the 2nd subject
- sub3 - name of the 3rd subject
- sub1_r - grade for the 1st subject obtained by student
- sub2_r - grade for the 2nd subject obtained by student
- sub3_r - grade for the 3rd subject obtained by student
- cgt_r - General knowledge marks
- grade - General knowledge grades
- ge_r - General English grades
- syllabus - has mentioned the student took the exam under which syllabus (old/new)
- gender - sex of the students
- age- age of the students

Dataset: <https://www.kaggle.com/datasets/sasikaamarasinghe/student-performance-gce-al-exam-2020-sri-lanka>

Problem Definition

Predict the overall Z-score (continuous) based on subject marks and background features (stream ,gender, age).

Objectives

- 1.To analyze students' academic performance by examining Z-score distribution and performance across streams.
- 2.To examine the effect of demographic factors such as age and gender on students' Z-scores.
- 3.To gain insights into the education system by comparing performances under the new and old syllabi and analyzing the relationship between the Common General Test and Z-scores.
- 4.To determine whether university eligibility differs by gender, age, syllabus type, and academic stream.

DATA PREPROCESSING & METHODOLOGY

1. Data cleaning

- In the census dataset, missing values were denoted by the symbol '-!'.
The variables island rank and district rank were excluded due to a high proportion of missing values and because they were not relevant to the study objectives.
- Missing values in the stream variable were imputed by grouping students according to their three main subjects and mapping the missing streams based on these groupings (logical imputation).
- Records with missing values in z-score and gender were removed to ensure data integrity.

```
Number of missing values per column (represented as '-!'):
stream          150514
Zscore          105249
district_rank   153254
island_rank     153254
gender          1484
dtype: int64
```

```
index          0
stream         0
Zscore        105249
al_year        0
sub1           0
sub1_r         0
sub2           0
sub2_r         0
sub3           0
sub3_r         0
cgt_r          0
ge_r           0
syllabus       0
birth_day      0
birth_month    0
birth_year     0
gender         1484
dtype: int64
```

2. Sampling

- A stratified random sampling technique was employed, with stream and syllabus as the stratification variables to ensure proportional representation.
- Using Cochran's formula, a sample size of 383 students was calculated and drawn from the cleaned dataset.

3. Variable Creation

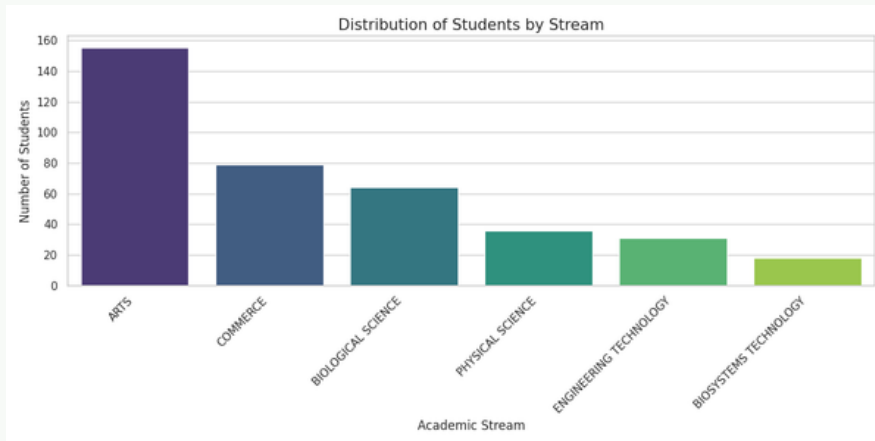
- A new binary variable was created to indicate eligibility for university entrance. A student was considered eligible if they obtained at least three 'S' grades in their main subjects.
- A new numerical variable 'age' was created using the 'al_year' and 'birth_year'.
- A new categorical variable was created as grade to convert cgt marks into grades

4. Analysis

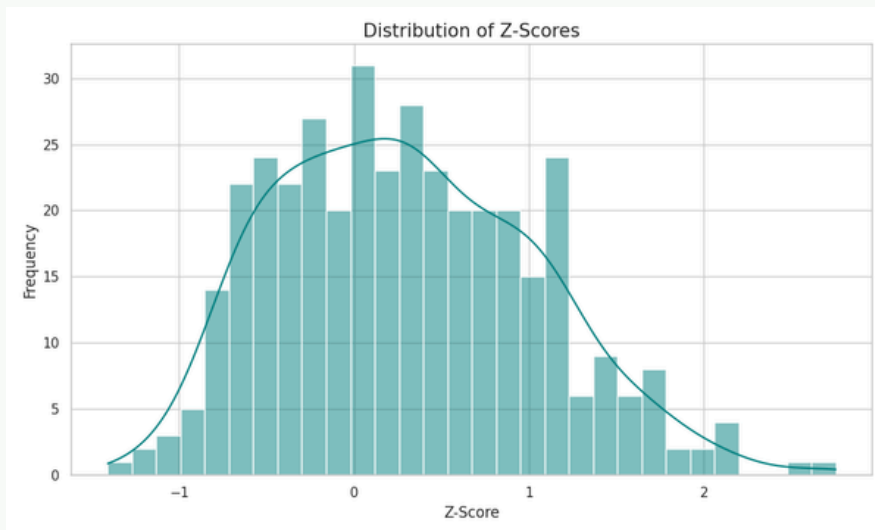
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Developed regression models for each specific objective.

EXPLORATORY DATA ANALYSIS

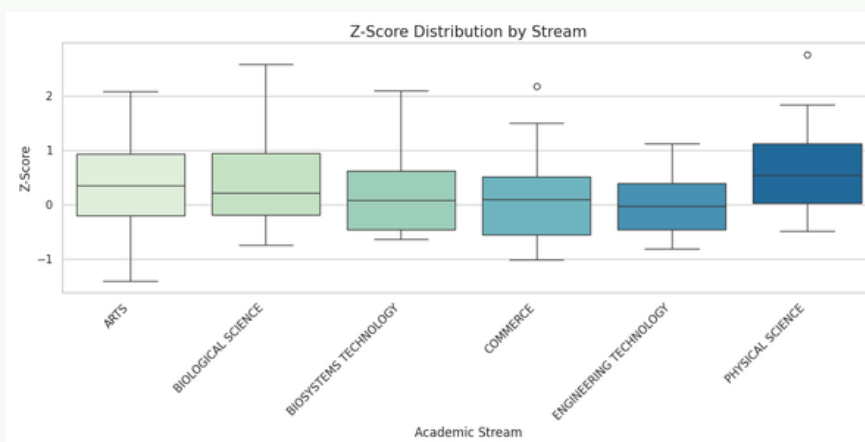
DISTRIBUTIONS OF STREAM AND Z-SCORE



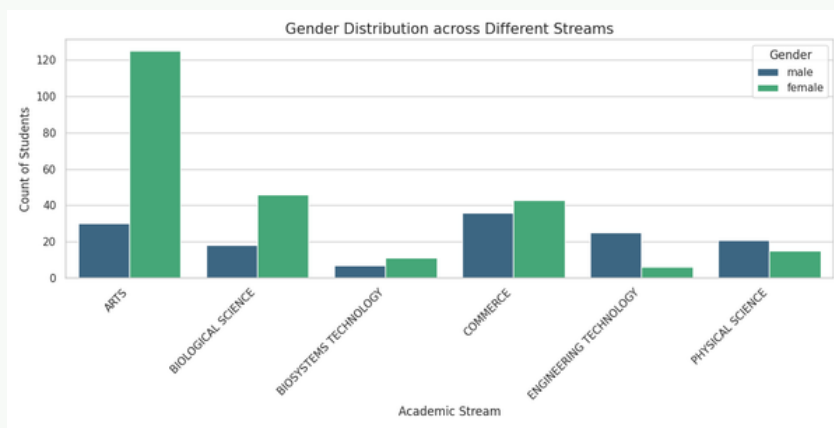
This is a bar chart drawn on the distribution of students by stream. It can be observed that the most number of students are from the Arts stream while the least number of students are from the Biosystems Technology stream.



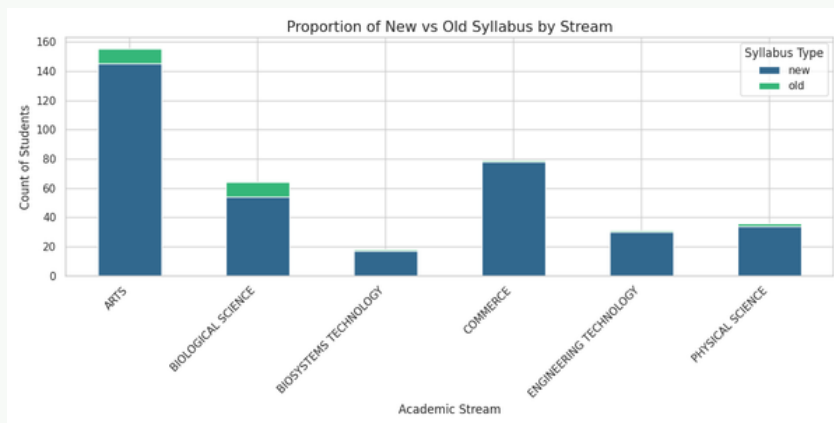
This is the histogram with a Kernel Density Estimate line plotted to observe the distribution of z-scores. It can be observed that the distribution is positively skewed, indicating that there are only a few students with extremely high z-scores. The mean is higher than the median of this distribution.



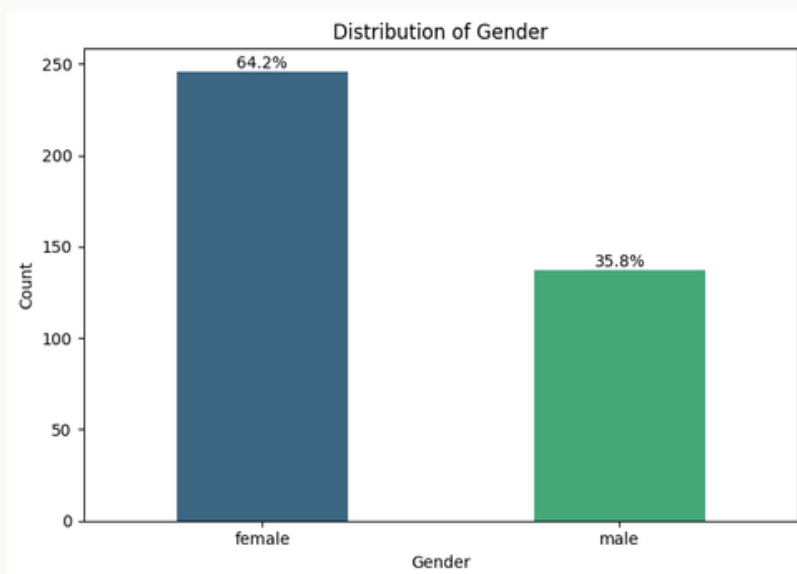
This is the boxplot plotted to observe distribution of z-score by each stream of study. It can be observed that the Physical Science stream has the highest median score indicating that this is the stream that students have achieved the highest results. Also, the Arts stream and the Biological Science stream have the widest ranges which indicates that there are both high and low achieving students mixed in these groups. There are a few outliers in the Commerce and Physical Science streams.



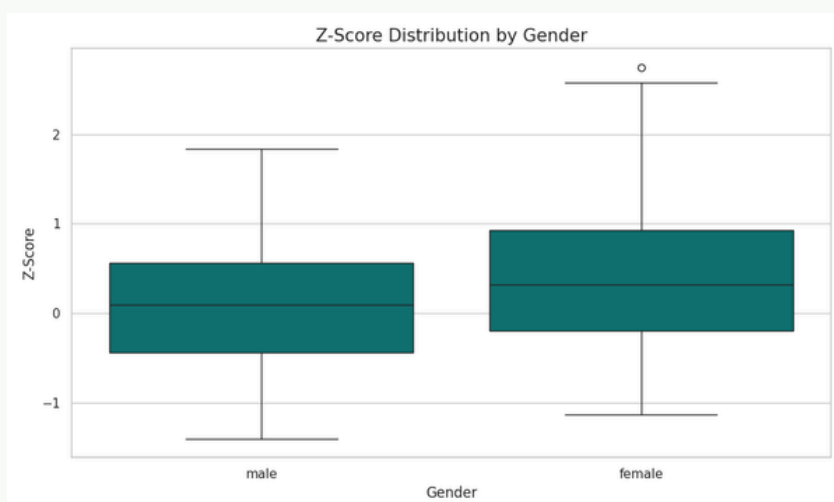
This is a grouped count plot plotted to observe the distribution of gender across streams. All other streams except for Engineering Technology and Physical Science have more females than males.



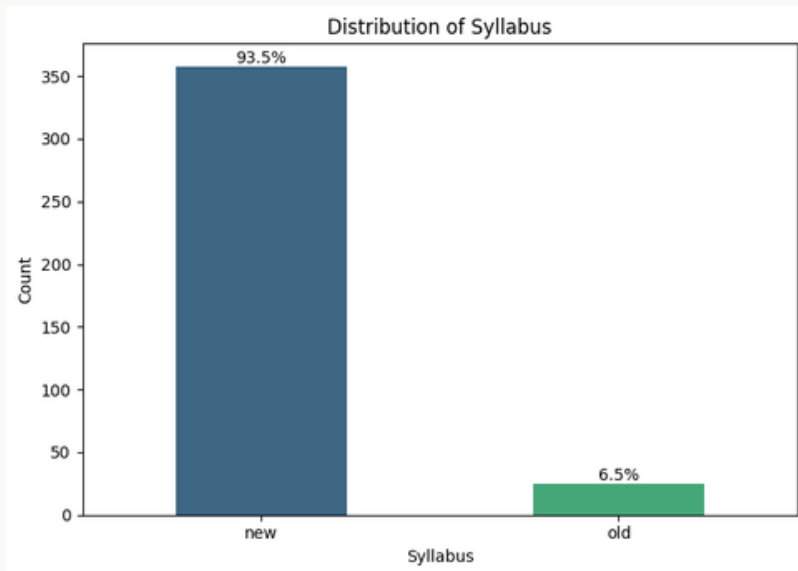
This stacked bar chart is to observe the proportions of the new and old syllabus by streams. It can be observed that Arts and Biological Science streams have a few old syllabus students while the other streams either have extremely few or no old syllabus students at all.



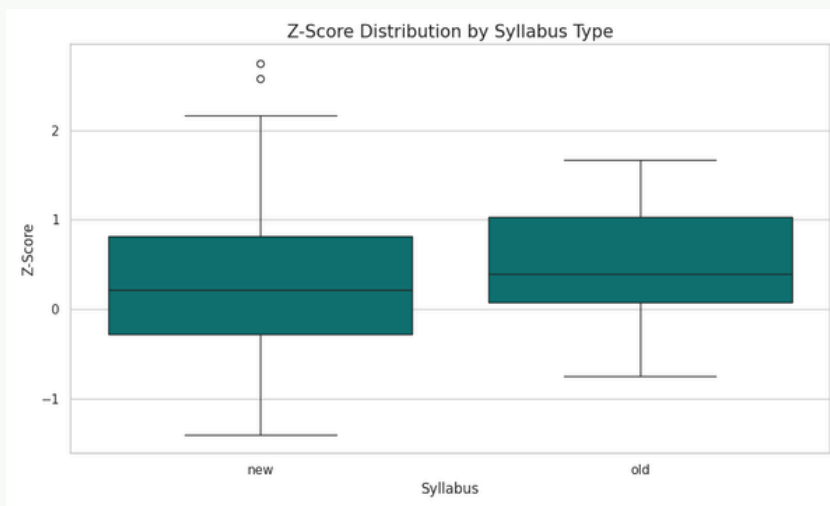
This graph shows the gender distribution of the students in the dataset. Female students make up the majority, accounting for about 64.2% of the total sample, while male students represent around 35.8%. This indicates that the dataset is female-dominated, with nearly two-thirds of the students being female.



This is the boxplot of z-score with gender. It can be observed that the median z-score of females is higher than men. Also, the range of spread of z-score is high in females than in men. This means that there is a higher range of z-score values for females than men. It can also be observed that there are some outliers in the -score distribution of females.

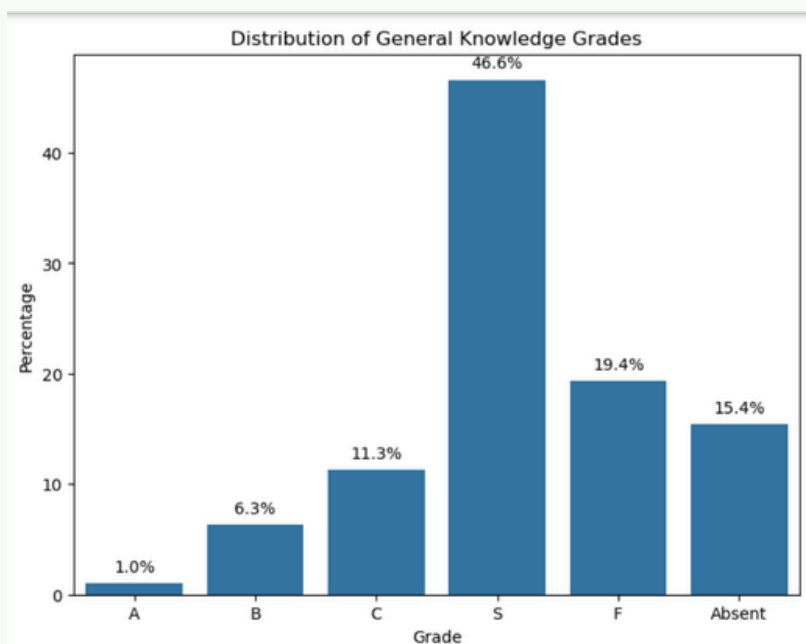


This graph shows the distribution of students by syllabus type. The vast majority of students, about 93.5%, follow the new syllabus, while only 6.5% are studying under the old syllabus. This indicates that the dataset is heavily dominated by students from the new syllabus, with very few students representing the old syllabus.

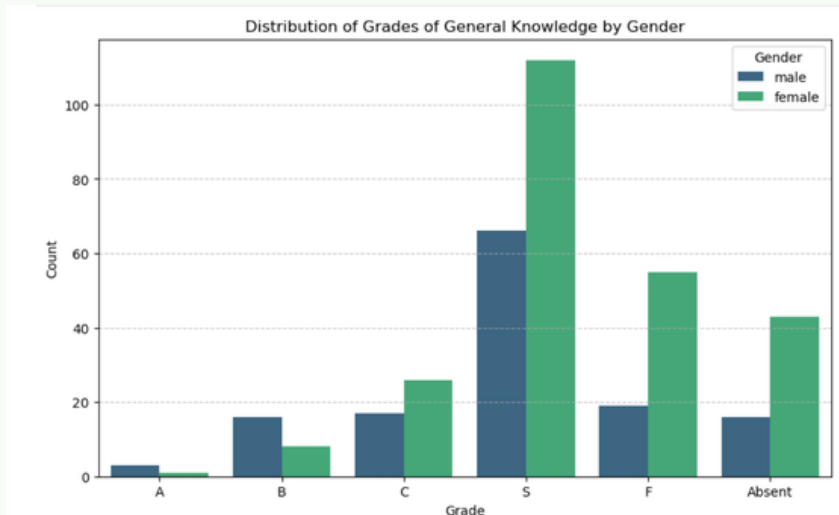


This is the distribution of z-score by syllabus type. The median of the old syllabus students is higher than the new syllabus students. The range of z-score of new syllabus students are higher than old syllabus students. There are a few outliers in the distribution of the new syllabus students.

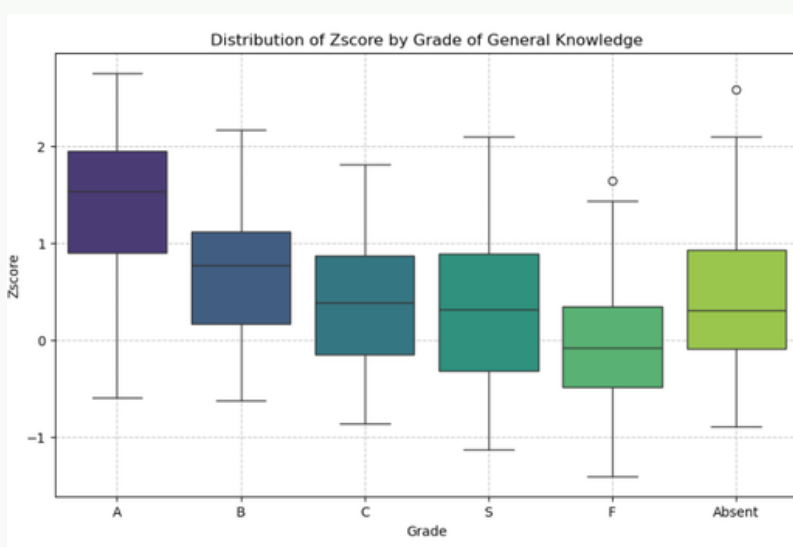
Distribution of the General Knowledge & General English Grades of Students



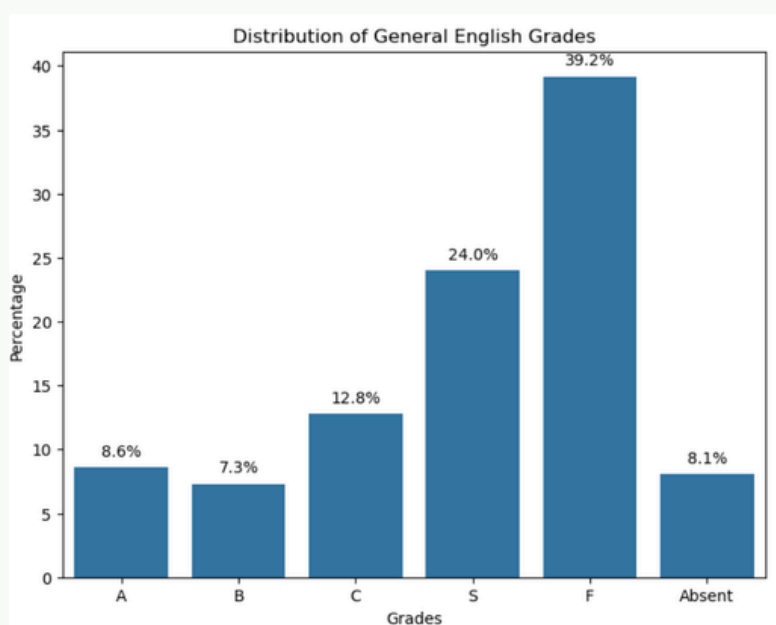
This bar plot illustrates how the students achieve grades for the General Knowledge Test. The majority of the students obtained 'S' passes for the general knowledge test. The fewest students obtained 'A' pass on the test. The percentage of 19.4% students failed the General Knowledge test in the 2020 A/L examination. Absent students were recorded as 15.4% for this test.



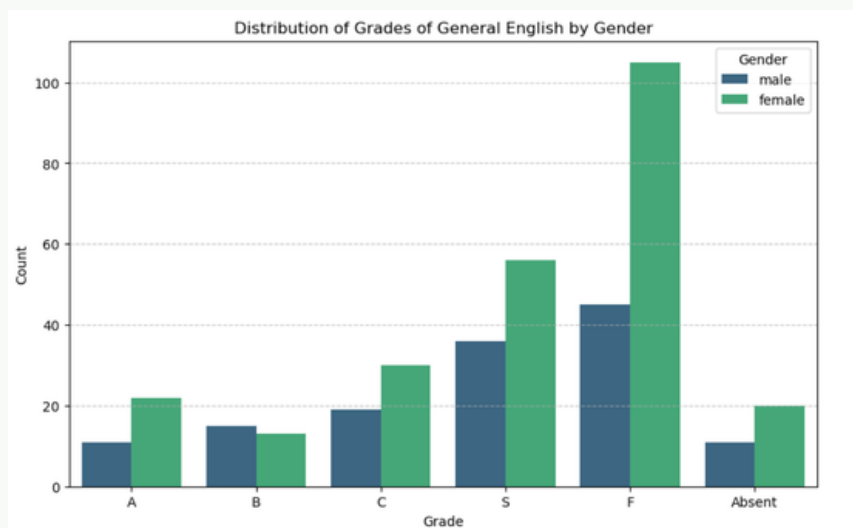
This graph describes the distribution of Grades of the General Knowledge test by gender. Most of the students who obtained 'S' and 'F' passes are female. When considering higher grades like 'A' and 'B', the male count is higher than the female count.



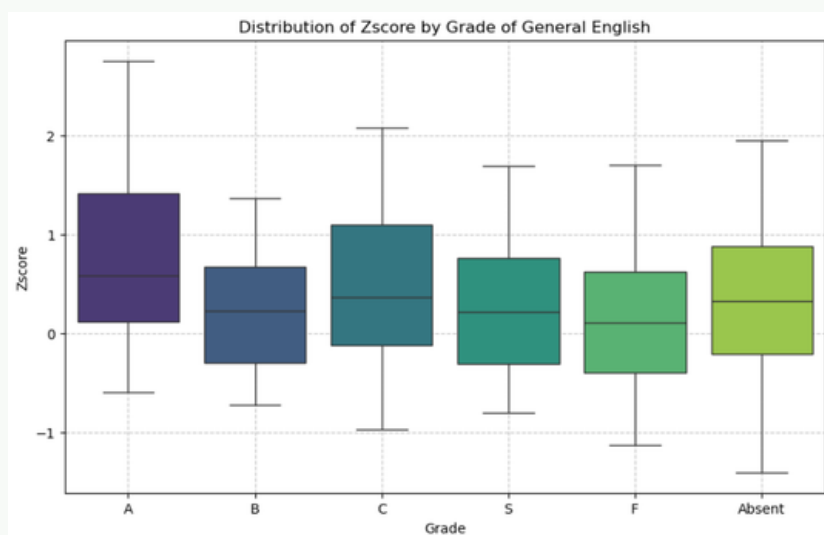
This representation illustrates the Distribution of Z scores by General Knowledge grades. This box plot clearly explains that there is a positive relationship between z-scores and General Knowledge grades. The students who got 'A' passes for the General Knowledge test have achieved a higher median z-score than students in other grades. The students who failed the General Knowledge test have achieved a minus z-score value as their median z-score.



This plot illustrates how students achieved grades for the General English test. According to this plot, the majority of students obtained lower grades. The percentage 39.2% of students who failed the test. Only 8.6% percent students obtained an 'A' pass for the test. The percentage of 8.1% students were absent for the test.



This plot describes how the grades of the General English test vary with the gender of the students. The plot explains that the majority of students who obtained lower grades are female. The number of females who got 'A' pass for the test is higher than that of males. Most of the failed students are female.

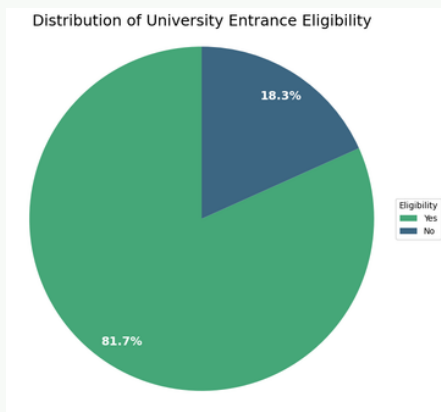


The box plot shows that students with higher General English grades generally have higher Z-scores. Grade A students perform the best on average, while performance gradually decreases C to S and F. Overall, better English grades are linked to better performance, but the relationship is not perfectly clear-cut.

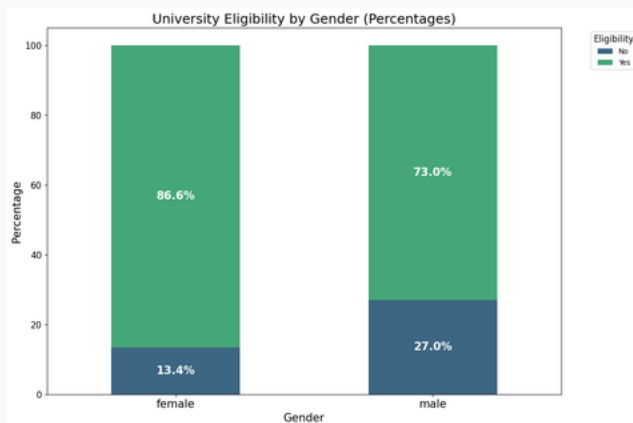
Distributions of University Eligibility

	Subject 1	Subject 2	Subject 3
A	28	34	39
B	50	63	82
C	113	132	122
S	155	136	125
F	37	18	15

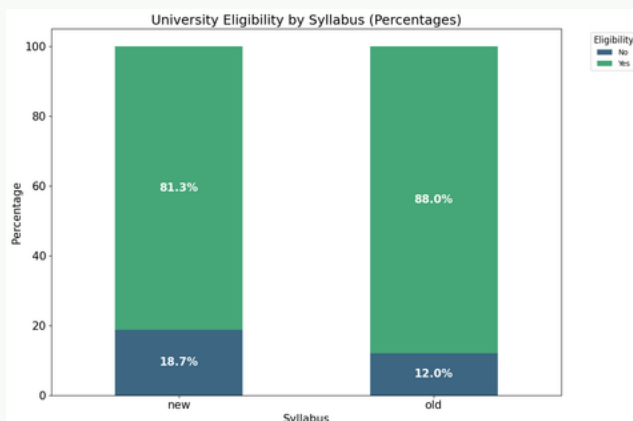
This table presents the frequency distribution of grades obtained for the three main Advanced Level subjects. The majority of students have obtained grades S and C across all three subjects, indicating that most candidates achieved at least the minimum pass requirement. Grade A frequencies are comparatively low, while fail (F) grades are present in all subjects but in smaller proportions.



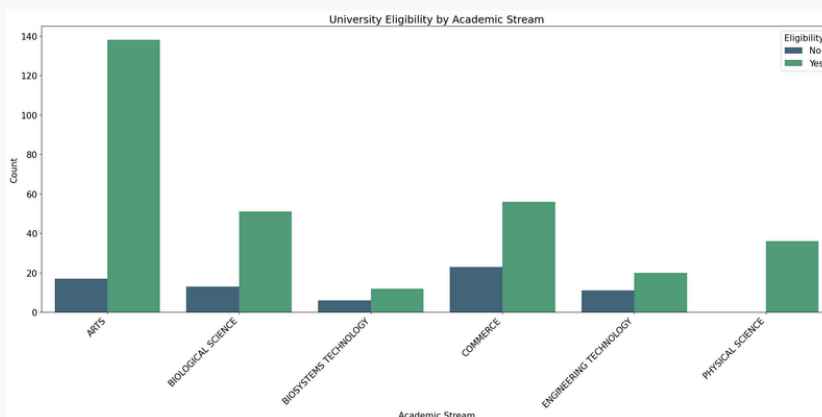
The pie chart illustrates the distribution of university entrance eligibility among students. It shows that 81.7% of the students are eligible for university admission, whereas 18.3% are not eligible.



This bar chart represents the university eligibility by the gender. From female students 86.6% are eligible for university entrance while from male students 73% are eligible for the university entrance.



The bar chart illustrates the distribution of university entrance eligibility among students, categorized by syllabus. It shows that 88.0% of students from the old syllabus are eligible for university admission, compared to 81.3% of students from the new syllabus.

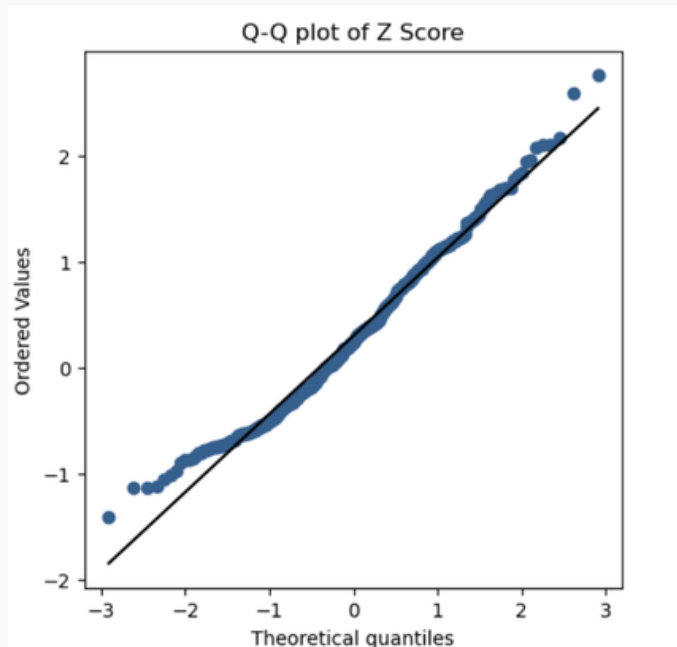


The bar chart illustrates the distribution of university entrance eligibility among students, categorized by their academic stream. It shows that in every academic stream, the number of students eligible for university admission exceeds those who are not eligible. In physical science stream all the students are eligible for the university entrance.

Objective 1

To analyze students' academic performance by examining Z-score distribution and performance across streams.

The Q-Q plot is showed that Z score data are approximately normally distributed.



Therefore parametric tests are used to further analysis.

Examine Z score across the different streams.

There are six streams: art, commerce, physical science, biological science, biosystems technology, and engineering technology. Examined whether the means of the z scores across streams are equal using a one-way ANOVA test. Firstly, used Levene's test on the variance of the z scores. According to that,

H0: Equal Variances are assumed

H1: Equal variances are not assumed

LeveneResult statistic=0.9301681506735054

pvalue=0.4612776163944048

P-value is greater than 0.05. So do not reject H0. Therefore, there is not enough evidence to say that equal variances are not assumed at 5% significance level.

Used test: One-way ANOVA was applied.

H0: Mean Z-scores are equal across all streams.

H1: At least one stream has a different mean Z-score.

Test Statistics

F-statistic: 4.4862

P-value: 0.0006

P-value is less than 0.05.

Statistical conclusion: Reject H0 at 5% significance level.

General Conclusion: Therefore, there is enough evidence to conclude that at least one stream has a different mean Z score at 5% significance level.

To identify which stream/s have different mean Z scores, the Tukey HSD Post-Hoc test was used.

```

--- Standalone Tukey HSD Post-Hoc Test Results ---
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1      group2      meandiff p-adj  lower  upper  reject
-----
          ARTS      BIOLOGICAL SCIENCE      0.058 0.9945 -0.2496 0.3656  False
          ARTS      BIOSYSTEMS TECHNOLOGY -0.1699 0.9348 -0.6854 0.3456  False
          ARTS      COMMERCE      -0.2822 0.0557 -0.5684 0.0039  False
          ARTS      ENGINEERING TECHNOLOGY -0.3412 0.1589 -0.7485 0.0661  False
          ARTS      PHYSICAL SCIENCE      0.2604 0.3748 -0.1226 0.6434  False
BIOLOGICAL SCIENCE BIOSYSTEMS TECHNOLOGY -0.2278 0.8455 -0.7801 0.3245  False
BIOLOGICAL SCIENCE COMMERCE      -0.3402 0.0598 -0.6884 0.0079  False
BIOLOGICAL SCIENCE ENGINEERING TECHNOLOGY -0.3992 0.1197 -0.8522 0.0538  False
BIOLOGICAL SCIENCE PHYSICAL SCIENCE      0.2025 0.7598 -0.2288 0.6338  False
BIOSYSTEMS TECHNOLOGY COMMERCE      -0.1124 0.9913 -0.6531 0.4283  False
BIOSYSTEMS TECHNOLOGY ENGINEERING TECHNOLOGY -0.1714 0.9674 -0.7849 0.4421  False
BIOSYSTEMS TECHNOLOGY PHYSICAL SCIENCE      0.4303 0.3094 -0.1673 1.0279  False
          COMMERCE ENGINEERING TECHNOLOGY -0.059 0.9989 -0.4977 0.3797  False
          COMMERCE PHYSICAL SCIENCE      0.5427 0.003  0.1264 0.959  True
ENGINEERING TECHNOLOGY PHYSICAL SCIENCE      0.6017 0.0097  0.0944 1.1089  True
=====

```

According to the results, the mean z score between physical science and commerce, and the mean z score between engineering technology and physical science, are significantly different from the others.

Then, regression model was built to quantify effect size from streams on z score.

$$z(i) = 0.3572 + 0.058 \cdot \text{Biological_science} - 0.1699 \cdot \text{Biosystem_technology} - 0.2822 \cdot \text{commerce} - 0.3412 \cdot \text{engineering_technology} + 0.2604 \cdot \text{physical_science}$$

Reference category is art stream

```

=====
OLS Regression Results
=====
Dep. Variable:      Zscore      R-squared:      0.056
Model:              OLS      Adj. R-squared:      0.044
Method:              Least Squares      F-statistic:      4.486
Date:                Sun, 25 Jan 2026      Prob (F-statistic):      0.000553
Time:                16:24:55      Log-Likelihood:      -416.05
No. Observations:    383      AIC:      844.1
Df Residuals:        377      BIC:      867.8
Df Model:            5
Covariance Type:     nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.3572	0.058	6.153	0.000	0.243	0.471
C(stream)[T.BIOLOGICAL SCIENCE]	0.0580	0.107	0.540	0.590	-0.153	0.269
C(stream)[T.BIOSYSTEMS TECHNOLOGY]	-0.1699	0.180	-0.944	0.346	-0.524	0.184
C(stream)[T.COMMERCE]	-0.2822	0.100	-2.825	0.005	-0.479	-0.086
C(stream)[T.ENGINEERING TECHNOLOGY]	-0.3412	0.142	-2.400	0.017	-0.621	-0.062
C(stream)[T.PHYSICAL SCIENCE]	0.2604	0.134	1.948	0.052	-0.002	0.523

```

=====
Omnibus:      9.955      Durbin-Watson:      1.958
Prob(Omnibus):      0.007      Jarque-Bera (JB):      9.727
Skew:         0.351      Prob(JB):      0.00772
Kurtosis:     2.660      Cond. No.      5.51
=====

```

As an example, physical science students have scored 0.2604 units higher than art students, and also commerce students have scored 0.2822 units lower than art students in the 2020 A/L Examination. Mean z score of art stream students is 0.3572.

Objective 2

To examine the effect of demographic factors such as age and gender on students' Z-scores.

Z-score differences between gender

H0: There is no significant difference in mean z-score values between genders

H1: There is a significant difference in mean z-score values between genders

Used test - Independent sample t-test

Test statistic:

t-statistic = -3.5707

p-value = 0.0004

- **Statistical conclusion**- Since the p-value=0.0004 is less than $\alpha=0.05$, we reject H0 at 5% level of significance.
- **General conclusion**- Therefore, we have enough evidence to conclude that there is a significant difference in the mean z-score values between genders.

Relationship between Age and Z-score

H0: There is no linear relationship between age and Z-score.

H1: There is a significant linear relationship between age and Z-score.

Used test - Pearson Correlation Analysis

Test statistic:

Pearson correlation coefficient (r) = 0.0244

p-value = 0.6341

- **Statistical conclusion:** Since the p-value=0.6321 is greater than $\alpha=0.05$, we do not reject H0 at 5% level of significance.
- **General conclusion:** Therefore, we have enough evidence to conclude that there is no linear relationship between age and Z-score

The Pearson correlation coefficient, 0.0244, shows that the relationship between age and z-score is a very weak or negligible, positive relationship.

Fitting a multiple linear regression model

‘Gender_encoded’ is the variable created by encoding Females as 1 and Males as 0.

```
--- Multiple Linear Regression Summary ---
                        OLS Regression Results
=====
Dep. Variable:          Zscore      R-squared:          0.033
Model:                  OLS         Adj. R-squared:       0.028
Method:                 Least Squares   F-statistic:       6.532
Date:                  Tue, 27 Jan 2026   Prob (F-statistic): 0.00163
Time:                  08:20:14         Log-Likelihood:    -420.65
No. Observations:      383             AIC:              847.3
Df Residuals:          380             BIC:              859.1
Df Model:              2
Covariance Type:       nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
const                -0.1809      0.522     -0.346     0.729     -1.208      0.846
age                   0.0154      0.027      0.579     0.563     -0.037      0.067
gender_encoded        0.2783      0.078      3.582     0.000      0.126      0.431
=====
Omnibus:              10.924   Durbin-Watson:       1.860
Prob(Omnibus):         0.004   Jarque-Bera (JB):    10.905
Skew:                  0.381   Prob(JB):            0.00428
Kurtosis:              2.677   Cond. No.           275.
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

But we can see that the p value of Age=0.563 is greater than alpha=0.05. Therefore, Age is not a significant variable. This indicates that the age of the students does not have an effect on the academic performance of the students

```
---
                        OLS Regression Results
=====
Dep. Variable:          Zscore      R-squared:          0.032
Model:                  OLS         Adj. R-squared:       0.030
Method:                 Least Squares   F-statistic:       12.75
Date:                  Thu, 29 Jan 2026   Prob (F-statistic): 0.000402
Time:                  15:26:02         Log-Likelihood:    -420.82
No. Observations:      383             AIC:              845.6
Df Residuals:          381             BIC:              853.5
Df Model:              1
Covariance Type:       nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
const                 0.1195      0.062      1.922     0.055     -0.003      0.242
gender_encoded        0.2771      0.078      3.571     0.000      0.125      0.430
=====
Omnibus:              10.907   Durbin-Watson:       1.854
Prob(Omnibus):         0.004   Jarque-Bera (JB):    10.777
Skew:                  0.375   Prob(JB):            0.00457
Kurtosis:              2.663   Cond. No.           3.10
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Therefore, our final model can be written as;

$$\text{Z-score} = 0.1195 + 0.2771(\text{Gender})$$

Interpretation of the Model:

The constant (0.1195): This is the predicted Z-score if both age and gender (encoded as 0) were zero. This serves as a baseline for the equation, even though it doesn't have a meaning, since age can't be zero.

Gender (0.2775): Since we encoded Female as 1 and Male as 0, this coefficient tells us that Female students are predicted to have a Z-score 0.2771 points higher than Male students, holding age constant.

Objective 3

To gain insights into the education system by comparing performances under the new and old syllabi and analyzing the relationship between the Common General Test and Z-scores.

Z-score difference between New and Old syllabus

H0 : There is no significant difference in mean z-scores between students following the new syllabus and the old syllabus

H1 : There is a significant difference in mean z-scores between students following the new syllabus and the old syllabus.

Used test - Independent sample t-test

Test statistic:

t-statistic = -0.9856

p-value = 0.3249

- **Statistical conclusion** - Since the p-value (0.3249) is greater than the significance level ($\alpha = 0.05$), we fail to reject H0 at the 5% significance level.
- **General conclusion** - There is no sufficient evidence to conclude that the syllabus type (new or old) has a significant effect on students' academic performance as measured by z-scores.

Relationship between CGT and Z-score

H0 : There is no linear relationship between CGT and z-scores

H1 : There is a linear relationship between CGT and z-scores

Used test - Pearson Correlation Analysis

Test statistic:

Pearson correlation coefficient (r) = 0.3070

p-value = 0.0000

- **Statistical conclusion** - The Pearson correlation coefficient indicates a weak positive relationship between CGT scores and Z-scores.
- **General conclusion** - There is evidence of a positive association between CGT performance and academic achievement, suggesting that students with higher CGT scores tend to obtain higher Z-scores.
- However, the relationship is very weak, indicating that CGT does not completely explain the variations in academic performance.

Fitting a multiple linear regression model

The new variable 'is_new_syllabus' was created by giving '1' if the student is from the new syllabus group and '0' if the student is from the old syllabus group.

```
--- Regression Summary ---
                        OLS Regression Results
=====
Dep. Variable:          Zscore      R-squared:          0.009
Model:                  OLS         Adj. R-squared:       0.004
Method:                 Least Squares   F-statistic:        1.813
Date:                  Thu, 29 Jan 2026   Prob (F-statistic):  0.165
Time:                  15:00:49         Log-Likelihood:     -425.30
No. Observations:      383             AIC:               856.6
Df Residuals:          380             BIC:               868.5
Df Model:              2
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                0.3416      0.159        2.148      0.032      0.029      0.654
cgt_r                0.0031      0.002        1.628      0.104     -0.001      0.007
is_new_syllabus      -0.1728      0.153       -1.128      0.260     -0.474      0.128
=====
Omnibus:              10.349   Durbin-Watson:       1.864
Prob(Omnibus):        0.006   Jarque-Bera (JB):    10.731
Skew:                 0.394   Prob(JB):            0.00467
Kurtosis:             2.775   Cond. No.            246.
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

This statistical analysis reveals that neither the syllabus type nor the Common General Test score has a significant impact on the Z-score.

Syllabus Comparison: The analysis showed no statistically significant difference in performance between students under the New Syllabus and the Old Syllabus ($p > 0.05$). This suggests that the Z-score calculation effectively normalizes results across different curricula, maintaining equity in the selection process.

Common General Test (CGT): The correlation between CGT scores and Z-scores was found to be very weak that it is insignificant. This confirms that the CGT assesses general aptitude and logical reasoning, which are distinct from the subject-specific academic success measured by the Z-score.

Objective 4

To determine whether university eligibility differs by gender, age, syllabus type, and academic stream.

University eligibility by gender

H0 : There is no association between gender and university eligibility

H1 : There is an association between gender and university eligibility

Used test - Chi square test

Chi-squared statistic: 9.9938

P-value: 0.0016

- **Statistical conclusion** - Since the p value is less than 0.05 we have to reject the null hypothesis at 5% significant level.
- **General conclusion** - We have enough evidence to say that there is an association between gender and university eligibility.

University eligibility by syllabus

H0 : There is no association between Syllabus (new/old) and university eligibility

H1 : There is an association between Syllabus (new/old) and university eligibility

Used test - Chi square test

Chi-squared statistic : 0.3275

P-value : 0.5671

- Statistical conclusion - Since the p value is greater than 0.05 we don't have enough evidence to reject the null hypothesis at 5% significant level.
- General conclusion - We don't have enough evidence to say that there is an association between syllabus and university eligibility.

University eligibility by stream

H0 : There is no association between stream and university eligibility

H1 : There is an association between stream and university eligibility

Used test - Chi square test

Chi-squared statistic: 28.8614

P-value: 0.0000

- **Statistical conclusion** - Since the p value is less than 0.05 we have to reject the null hypothesis at 5% significant level.
- **General conclusion** - We have enough evidence to say that there is an association between stream and university eligibility.

Fitted a logistic model to find the significant factors associated with the university eligibility.

In the stratified sample, 100% of the students in the Physical Science stream were eligible for university entrance (zero "F" grades) which led to perfect separation. To solve this, the Physical Science and Biological Science streams were merged into a single "Science" category.

```

... Optimization terminated successfully.
      Current function value: 0.442009
      Iterations 6

      Logit Regression Results
=====
Dep. Variable:    eligible_for_university_entrance    No. Observations:    383
Model:            Logit                               Df Residuals:        375
Method:           MLE                               Df Model:            7
Date:             Thu, 29 Jan 2026                   Pseudo R-squ.:       0.07056
Time:            12:29:19                             Log-Likelihood:      -169.29
converged:        True                               LL-Null:             -182.14
Covariance Type:  nonrobust                          LLR p-value:         0.0005690
=====
               coef    std err          z      P>|z|      [0.025    0.975]
-----
const                2.6551      1.958      1.356     0.175     -1.183     6.493
gender_male          -0.6011      0.293     -2.050     0.040     -1.176     -0.026
syllabus_old         0.3050      0.690      0.442     0.659     -1.048     1.658
stream_grouped_BIOSYSTEMS TECHNOLOGY -1.2942      0.569     -2.276     0.023     -2.409     -0.180
stream_grouped_COMMERCE -1.0478      0.371     -2.827     0.005     -1.774     -0.321
stream_grouped_ENGINEERING TECHNOLOGY -1.1445      0.490     -2.337     0.019     -2.104     -0.184
stream_grouped_Science -0.0833      0.402     -0.207     0.836     -0.872     0.706
age                 -0.0223      0.099     -0.225     0.822     -0.217     0.172
=====

```

After removing the insignificant variables following model was obtained.

```

Optimization terminated successfully.
      Current function value: 0.442281
      Iterations 6

      Logit Regression Results
=====
Dep. Variable:    eligible_for_university_entrance    No. Observations:    383
Model:            Logit                               Df Residuals:        377
Method:           MLE                               Df Model:            5
Date:             Thu, 29 Jan 2026                   Pseudo R-squ.:       0.06999
Time:            12:29:19                             Log-Likelihood:      -169.39
converged:        True                               LL-Null:             -182.14
Covariance Type:  nonrobust                          LLR p-value:         0.0001118
=====
               coef    std err          z      P>|z|      [0.025    0.975]
-----
const                2.2347      0.270      8.274     0.000      1.705      2.764
gender_male          -0.6050      0.293     -2.068     0.039     -1.178     -0.032
stream_grouped_BIOSYSTEMS TECHNOLOGY -1.2924      0.569     -2.271     0.023     -2.408     -0.177
stream_grouped_COMMERCE -1.0504      0.366     -2.868     0.004     -1.768     -0.332
stream_grouped_ENGINEERING TECHNOLOGY -1.1396      0.486     -2.343     0.019     -2.093     -0.186
stream_grouped_Science -0.0652      0.400     -0.163     0.871     -0.850      0.719
=====

```

Although the logistic regression model is statistically significant, the pseudo R² value (6.9%) indicates that the model has limited explanatory power.

$\ln(p/1-p) = 2.235 - 0.605(\text{Male}) - 1.292(\text{Biosystems}) - 1.050(\text{Commerce}) - 1.140(\text{Eng Tech}) - 0.065(\text{Science})$

Interpretations

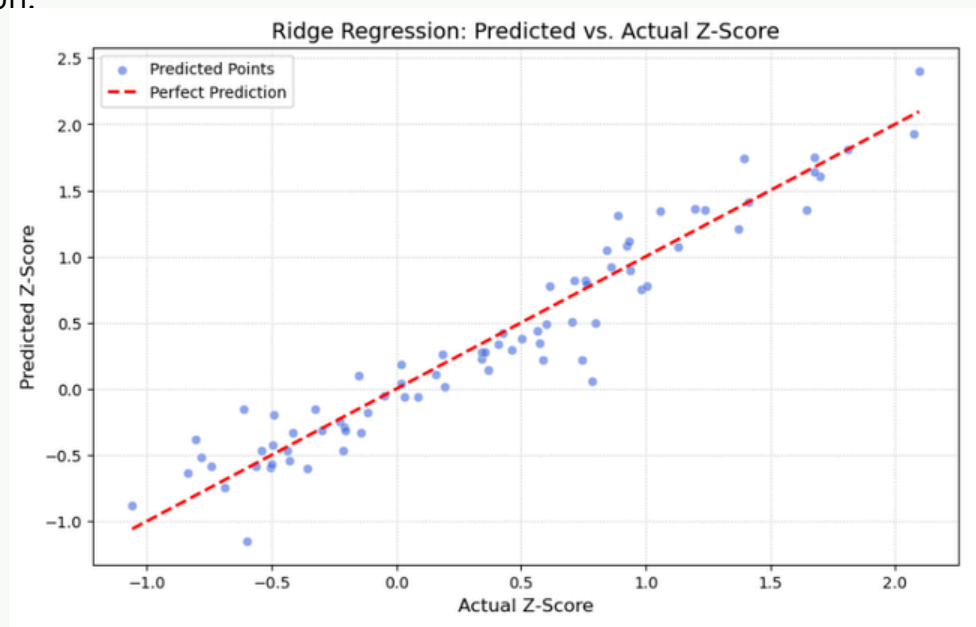
- Males have about 45% lower odds of the outcome compared to females, holding other variables constant.
- Commerce students have about 65% lower odds than Art students.
- Biosystems students have about 73% lower odds than Art students.
- Engineering Technology students have about 68% lower odds than Art students.
- Science students have only about 6% lower odds than Art students

Predict the overall Z-score (continuous) based on subject marks and background features (stream ,gender, age).

Model	R ²	MAE	Hit Rate (+/- 0.2)
Ridge Regression	92.19%	0.1644	71.43%
Decision Tree	82.46%	0.2534	46.75%
Random Forest	91.13%	0.1833	59.74%
Gradient Boosting	92.96%	0.1638	64.94%
KNN	84.10%	0.2521	44.16%

While Gradient Boosting had a slightly higher accuracy score (92.96%), Ridge Regression was better at "Hitting the Mark." Its Hit Rate was 71.43%, meaning it was much more likely to predict a score within 0.2 points of the actual result.

Therefore, our analysis shows that Ridge Regression is the best model for predicting student Z-scores. Ridge Regression achieved an R² value of 0.922, indicating that approximately 92.2% of the variance in Z-scores is explained by the model. So, it can predict a student's results very closely using just their grades and basic background information.



The Predicted vs. Actual plot for the Ridge Regression model confirms its high reliability and precision.

Conclusion

This research provided a comprehensive analysis of the factors influencing student performance in the 2020 GCE Advanced Level examination in Sri Lanka. By employing a stratified random sampling technique and developing various predictive and inferential models, several key insights were uncovered.

Academic Achievement and Streams:

The analysis of Z-score distributions revealed that students in the Physical Science stream achieved the highest median results. While the Arts stream had the highest number of students, it also showed a wide range of performance, indicating a mix of high and low achievers.

Impact of Demographic Factors:

Gender emerged as a significant predictor of academic success. Female students not only made up the majority of the student population (64.2%) but also achieved a higher median Z-score compared to the male students.

Syllabus and General Testing:

The study found that while the majority of students followed the new syllabus, there is no difference between z scores of the old and new syllabi. Furthermore, no relationship was established between performance in the General Knowledge test and overall Z-scores.

University Eligibility:

The investigation into university entrance requirements highlighted significant disparities across different streams and demographic groups.

Predictive Model:

Through machine learning, the Gradient Boosting and Ridge Regression models proved to be highly effective, explaining nearly 92% of the variance in student Z-scores. Furthermore, the ridge regression model achieved the highest hit rate and R^2 values and was the best-performing model.